A Multiobjective Method Leveraging Spatial–Spectral Relationship for Hyperspectral Unmixing

Erfeng Liu, Zikai Wu, and Hongjuan Zhang

***Abstract*— Free of tuning regularization parameters, sparse unmixing based on multiobjective methods has become increasingly popular for the hyperspectral image in recent years. Moreover, inherent signatures of a hyperspectral image have been exploited in various single objective-based methods and proved relevant for improving unmixing performance. However, their utilizations in multiobjective optimization are still scarce. With the overarching goal of exploiting the spatial signature in an explicit way for hyperspectral unmixing, this work proposes a multiobjective method leveraging spatial spectral relationship for hyperspectral unmixing [graph-based multiobjective optimization method for sparse unmixing (GMoSU)]. First, a multiobjective sparse unmixing model based on spatial signatures encoded by the graph Laplacian is put forward. Then, to solve this model efficiently, an improved Tchebycheff decomposition approach and a novel local recombination strategy are rationally proposed, both of them and an operation of encoding the solution as a binary vector are plugged into the framework of multiobjective evolutionary algorithm based on decomposition (MOEA/D). Theoretically, the improved Tchebycheff formula formed by introducing a mixed spectral similarity metric enables the Pareto-optimal front to converge to a single solution exactly. Encoding the solution as a binary vector could help effectively address the endmember selection problem. The novel local recombination strategy that an individual is updated through recombining with another individual selected randomly in its neighborhood could balance the diversity and convergence of population further. Finally, comprehensive comparison experiments are conducted on synthetic and real datasets, which verify the theoretical advantages and effectiveness of the proposed GMoSU even under heavy noise.**

***Index Terms*— Hyperspectral image, multiobjective optimization, sparse unmixing, spatial information.**

I. Introduction

BENEFITING from the development of imaging spectromtery technology, hyperspectral images have been widely used in the fields of scientific research, geological exploration, and military [1], [2], [3]. Due to the higher spectral resolution of a sensor, hyperspectral images could reflect small differences in characteristic wavelengths of ground objects [4]. However, both the low spatial resolution and the materials combining from distinct pure substances could generate many mixed pixels [1], which will impact the application of hyperspectral imaging [1], [2], [3], [5]. Therefore, hyperspectral unmixing has become essential for getting a comprehensive understanding of the feature information in remote sensing images.

In general, unmixing includes two parts: endmember extraction and abundance inversion, which aim to extract the spectra of pure materials and to estimate their corresponding fractions from a mixed pixel, respectively [1]. At present, many kinds of related methods have been developed, for example, geometrical-based unmixing methods [6], [7], [8], [9], statistical-based methods [10], [11], [12], [13], nonnegative matrix factorization (NMF)-based methods [14], [15], [16], [17], sparse unmixing [18], [19], [20], [21], [22], [23], data driving-based methods [24], [25], [26], [27], and nonlinear mixing model-based methods [28], [29], [30], [31]. Note that majority of them are based on the linear mixing model (LMM) due to its low complexity and its feasibility in most real scenarios [4]. Specifically, geometrical-based unmixing method usually finds the vertices of a simplex that can enclose all the pixels. It performs well under the pure pixel assumption, but this assumption is difficult to hold in reality [2]. The statistical-based method generally uses parameter estimation techniques to solve the unmixing problems, which is suitable for unmixing highly mixed spectral data, but the computational complexity is higher [32]. The NMF-based method decomposes hyperspectral image into endmember matrix and abundance matrix. However, the virtual endmembers obtained by NMF have no physical meaning and are difficult to match with real substance spectra [33]. In recent years, sparse unmixing has received widespread attention. It linearly combines a small number of spectral signatures from a prior spectral library in semi-supervised manner, which does not need pure pixel assumption; the extracted endmembers have real physical meaning; and the computational complexity is acceptable [5]. The formulation of LMM can be written as

$$Y = AX + E \tag{1}$$

where $Y = [y_1, y_2, \ldots, y_N] \in R^{L\times N}$ is a whole hyperspectral image with L bands and N pixels, $A \in R^{L\times M}$ is a spectral library with M pure materials, $X = [x_1, x_2, \ldots, x_N] \in$

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$R^{M\times N}$ is a fractional abundance matrix, and $E \in R^{L\times N}$ is a noise matrix. Due to the physical meaning of fractional abundance, the fractional abundance vector comprising matrix X is subject to two constrains: the abundance nonnegativity constraint (ANC) and the abundance sum-to-one constraint (ASC), that is, $x \geq 0$ and $1^T x = 1$. However, literature [5] has pointed out that ASC is a little too strong and could give an automatic imposition to generalized ASC. Therefore, in this work, only ANC is imposed on fractional abundance vector. The optimization problem of sparse unmixing can be written as

$$\begin{aligned} \min_{X\geq 0} \quad & \|X\|_{\text{row}-0} \\ \text{s.t.} \quad & \|Y - AX\|_F \leq \delta \end{aligned} \tag{2}$$

where $\|X\|_{\text{row}-0}$ denotes the number of nonzero rows in X, namely the sparsity, $\| * \|_F$ is the Frobenius norm of the matrix, and δ denotes unmixing residuals generally caused by noise. However, sparse unmixing is a nonconvex and NP-hard combinational problem that is L_0-norm-based [34]. Researchers tackled this problem mainly in two methods: the convex relaxation method and the greedy algorithm. The convex relaxation method mostly replaces the L_0-norm with the L_1-norm [18], [19], [20] or L_p $(0 < p < 1)$-norm penalty term [35], [36], [37]. This kind of method can only provide a limited approximation for the L_0-norm, which may lead to loss of the original solution space information and causes unmixing so that some errors occur. In comparison, the greedy algorithm tries to deal with the L_0-norm directly and to find the optimal solution by iterative calculation manner [21], [38], [39], [40]. Nonetheless, the greedy algorithm is very sensitive to parameter settings, for example, the number of endmembers becomes unstable when different parameter inputting.

The traditional spectral unmixing methods discussed earlier properly belong to regularized optimization methods, which usually solves a single-objective optimization problem containing weighted regularized terms. Therefore, it is necessary to select appropriate parameters for regularized term in order to obtain better results [41]. However, due to the lack of information about the solutions, the selection of regularization parameters becomes very difficult [42]. Recently, many researchers have proposed various multiobjective optimization methods based on L_0-norm for solving sparse unmixing problems [43], [44], [45], [46], [47], [48], [49]. Xu and Shi [43] proposed a bi-objective-based sparse unmixing algorithm, in which the elitist nondominated sorting genetic algorithm (NSGA-II) is used successfully. However, its final solution excessively depends on the manual selection. In order to solve the aforementioned problem, a method in [44] promoted the convergence of the population by incorporating the spectral information divergence (SID) regularization to the Tchebycheff decomposition. However, the performance of this method will weaken in high noise. Following this work, literature [45] used a distributed estimation algorithm to guide population evolution and a classification-based strategy to generate new individuals in the population, improving the robustness for noise to a certain extent. Furthermore, Xu et al. [46] regarded the pruning mapping function as one of the optimization objectives, which effectively compresses the search space, but makes the computation time longer, since unmixing and pruning are performed simultaneously in iterations. In addition, Gong et al. [47] proposed a multiobjective cooperative coevolutionary algorithm, which guides population evolution through cooperation between individuals with the knee solution. Jiang et al. [48] divided endmember selection and abundance estimation into two stages, and established a bi-objective optimization problem in each stage. Based on [48], both the Euclidean distance (ED) and the spectral angle are incorporated for endmember selection in [49], which simultaneously considers the shape and amplitude discrepancy of spectra signals.

In summary, the multiobjective optimization method for sparse unmixing has its advantages and disadvantages. Specifically, it can directly treat the regularized terms as one of the optimization objectives, thus avoiding the selection of regularization parameters [50] and enhancing the robustness of the unmixing method. However, the multiobjective optimization problem (MOP) is solved mainly depending on the heuristic algorithm, such as particle swarm optimization (PSO) [51], NSGA-II [52], and multiobjective evolutionary algorithm based on decomposition (MOEA/D) [53], which needs to improve the species diversity for escaping from local optimality. To address the problem, some strategies have been proposed successfully, for example, Gaussian perturbation operator [54], adaptive adjusting mutation probability [55], and diversity-enhanced subset selection [56]. However, these methods do not take the interaction of information between individuals into account fully, which will underutilize the search space of the problem. Therefore, an effective alternative is employing mutation and recombination simultaneously for generating new solutions [57], [58]. Meanwhile, a Pareto-optimal front by solving MOPs will also be obtained. Different from a single solution, the Pareto-optimal front contains many nondominated solutions [59], the question that remains is how to select the final solution. At present, some tactics to this problem have been proposed, for example, Xu et al. [60] set the reference point as the final solution, in which the reference point corresponds to the real individual in MOEA/D. Jiang et al. consider the knee point [61] in Pareto-optimal front as the final solution [49]. Although these approaches perform better to some extent, they sometimes obtain wrong solutions or even multiple solutions in practice. Therefore, it is necessary to consider other available information, such as the spectral characterization in the population, for selecting the optimal single solution.

In addition, hyperspectral images not only contain high spectral resolution but also involve rich spatial distribution information of ground objects. Therefore, combining spatial information with spectral information will improve the unmixing accuracy to some extent [62]. For example, in [63], the morphological operations were first used to extract endmembers, which is considered as one of the earliest methods to consider the spatial spectral information in unmixing. Literature [64] embedded a total variation (TV) spatial regularization term to sparse unmixing, which characterizes the spatial relationship between pixels and neighboring pixels.

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Recently, Zhang et al. [65] proposed incorporating spatial and spectral weighting factor into the NMF framework to capture the piecewise smooth structure of the data. Chen et al. [66] proposed combining superpixel segmentation and structural sparsity to improve the performance of the traditional spatial regularization-based unmixing methods. However, since adjacent pixels in the scene should have similar material composition and similar fractional abundance [2], [32], a graph could be constructed, in which each node represents a pixel, and pairwise similarities are connected by edges. Under the condition that the spatial information can be considered by the graph Laplacian, some unmixing methods based on it have been proposed. For example, Tang et al. [21] incorporated the regularization term based on the graph Laplacian to obtain more accurate endmembers, which exploited the spatial–contextual coherence. In [67], a manifold regularized NMF method was proposed, which could keep the link between the hyperspectral data and the estimated abundance. Literature [68] introduced the graph Laplacian into sparse unmixing for promoting the abundance estimation smoothness. The graph Laplacian could provide a low-dimensional expression of the data while preserving the similarity between the original data [69], which has been applied in many fields successfully, such as signal processing [70], clustering problem [71], and deep learning [72].

As discussed earlier, spatial information has been successfully used for hyperspectral unmixing based on the traditional regularized optimization method. However, utilizing it in the multiobjective optimization framework is still less touched. Therefore, this work will design a multiobjective optimization unmixing method which harnesses the spatial information of hyperspectral images by the graph Laplacian.

On the one hand, inspired by the SMoSU, the binary code strategy is also applied for spectral signatures in the library. On the other hand, in order to overcome the limitation of the bitwise flipping strategy adopted in the SMoSU, we propose a local recombination strategy, in which an individual is updated by recombining it with another individual selected randomly in its neighborhood. This strategy makes it much easier to update solutions with any number of bits and to achieve a better exploration in the solution space.

Moreover, the spectral characterization, such as spectral amplitude, angle, and information entropy, could be reflected fully by the measures such as ED, spectral angle metric (SAM), and SID [73]. Motivated by the relevance and complementarity of ED, SAM, and SID, we will design a mixed spectral similarity metric based on the Tchebycheff formula in MOEA/D to help us selecting the final solution. Different from SMoSU that only employ the SID of hyperspectral data, the mixed metric combines the ED, SAM, and SID. Theoretically, the proposed mixed metric could enforce the solutions of the Pareto-optimal front to converge to a single solution more exactly.

In summary, the major contributions of graph-based multiobjective optimization method for sparse unmixing (GMoSU) are threefold.

1) A GMoSU is proposed, in which the graph Laplacian is introduced for effectively leveraging the spatial information of hyperspectral images.

2) A local recombination strategy is proposed and plugged it into the MOEA/D framework, which could both improve the diversity of species and mitigate the influence of large dictionary size in sparse unmixing.

3) A mixed spectral similarity metric is introduced for addressing the problem of final solution selection based on the Tchebycheff formula in MOEA/D, which could enforce the solutions of the Pareto-optimal front to converge to a single solution.

II. Related Work

In this section, we will review some related works.

A. Multiobjective Optimization Problem

An MOP can be described as [59]

$$\begin{aligned} \min \quad & F(x) = [f_1(x),\ f_2(x), \ldots, f_n(x)]^T \\ \text{s.t.} \quad & g_i(x) \le 0, \quad i = 1, 2, \ldots, k \\ & h_j(x) = 0, \quad j = 1, 2, \ldots, m \end{aligned} \tag{3}$$

where x is the decision variable, $f_1(x), f_2(x), \ldots, f_n(x)$ are objective functions that need to be solved, and $g_i(x)$ and $h_j(x)$ are constraint functions. Since these objective functions usually contradict each other, the aim of most multiobjective optimization is finding a set of best tradeoffs other than a best solution, and some definitions about MOP should be introduced as discussed in [50].

1) Pareto-Dominance: Assuming that feasible solution x_1 dominates x_2, denoted by $x_1 \prec x_2$, if and only if

$$\begin{aligned} & \forall i = 1, 2, \ldots, n, \quad f_i(x_1) \le f_i(x_2) \\ & \exists i = 1, 2, \ldots, n, \quad f_i(x_1) < f_i(x_2). \end{aligned} \tag{4}$$

2) Pareto-Optimal Solution: The solution x^* is called the Pareto-optimal solution or Pareto optimality if and only if there are no other feasible solutions that dominate x^*, and then $F(x^*)$ is called the Pareto optimal vector.

3) Pareto-Optimal Set: The set of all Pareto-optimal solutions of MOP is called the Pareto-optimal set.

4) Pareto-Optimal Front: All objective function values corresponding to the Pareto-optimal set are called Pareto-optimal front.

B. Tchebycheff Approach of MOEA/D

The MOEA/D method combines mathematical programming methods with evolutionary algorithms by decomposing the MOP into many single-objective optimization subproblems based on the aggregation function, then using population evolution method to approximate the optimal solution set of the original problem, for example, weighted sum approach, Tchebycheff approach, and Boundary Intersection Approach. In this section, we will introduce the Tchebycheff approach, which is a common and effective decomposition method. Assuming the population consists of n individuals, an MOP could be decomposed into n scalar optimization problems by the Tchebycheff formula, the ith subproblem can be written as

$$\begin{aligned} \min_x \quad & g_i^{te}\left(x|\lambda_i, z^*\right) = \max\{\lambda_i |f_i(x) - z_i^*|\} \\ \text{s.t.} \quad & x \in \Omega \end{aligned} \tag{5}$$

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where $z^* = (z_1^*, z_2^*, \ldots, z_n^*)^T$ is the reference (or ideal) point, $z_i^* = \min\{f_i(x)|x \in \Omega, i = 1, 2, \ldots, n\}$ and $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_n)^T$ is the weight vector, where λ_i corresponds to the ith subproblem. Note that there exists a neighborhood range for each λ^i, which consists of all the sub-problems with the weight vectors from the neighborhood of λ^i, thus optimizing the ith subproblem will use the information from other subproblems only in this neighborhood. Finally, an approximate Pareto-optimal set of MOP could be obtained, which is composed of all optimal solutions of all subproblems.

C. PORSS

The Pareto optimization algorithm with recombination for subset selection (PORSS) has been proposed in [74], which employs recombination and mutation simultaneously to generate new solutions. Its subset selection problem can be described as

$$\begin{aligned} &\min_{S \subseteq V} \quad f(S) \\ &\text{s.t.} \quad |S| \leq K. \end{aligned} \tag{6}$$

The aim of PORSS is to find the optimal solution of (6) by reformulating it into an MOP, which can be described as

$$\min_{S \in \{0,1\}^n} \quad (f_1(S), f_2(S)) \tag{7}$$

where

$$\begin{aligned} f_1(S) &= \begin{cases} +\infty, & |S| = 0 \quad or \quad \geq 2K \\ f(S), & \text{otherwise} \end{cases} \\ f_2(S) &= |S|. \end{aligned} \tag{8}$$

Here f is a criterion function, such as mean squared error. $S = [s_1, s_2, \ldots, s_n]$ is a binary vector, where $|S|$ is the number of nonzero element of S, i.e., the cardinality of S. $K \geq 0$ controls the sparsity of S. Given a library $A = [a_1, a_2, \ldots, a_n]$, we select at most K atoms of A by S, if a_i is selected, then $s_i = 1$, else $s_i = 0$. The set of f_1 to $+\infty$ is to exclude infeasible solutions.

PORSS employs evolutionary algorithms based on the Pareto domination relationship, in which recombination and bitwise mutation are used for generate new solutions, which can be described as follows.

1) One-Point Recombination: We randomly select two individuals x and y from the current population. Then, we randomly choose $i \in \{1, 2, \ldots, n\}$ and exchange the first i bits of x and y to generate x' and y'.

2) Uniform Recombination: We exchange all corresponding bits of x' and y' in probability $1/2$ to generate x'' and y''.

3) Bitwise Mutation: Flip each bit of x'' and y'' in probability $1/n$, where n is the encoding length of solutions. Compared to POSS [75] in that without recombination, PORSS performs better when the library is with a large number of atoms. The recombination strategy is more likely to generate diverse solutions and escape the local optima [74].

D. SMoSU

The unmixing method of SMoSU was developed by the inspiration of POSS, the formulation of it is

$$\min_{s \in \{0,1\}^m} \quad F(s) = [f_1(s), f_2(s)] \tag{9}$$

where

$$\begin{aligned} f_1(s) &= \begin{cases} +\infty, & \|s\|_1 = 0 \text{ or } \geq 2\text{k} \\ \|Y - A_s X_s\|_F, & \text{Otherwise} \end{cases} \\ f_2(s) &= |\|s\|_1 - k|. \end{aligned} \tag{10}$$

Here s is the binary decision variable, in which coding length m equals to the number of spectra in the library. However, A_s denotes the endmember matrix composed of the selected endmember, and X_s is the corresponding abundance matrix. Additionally, k refers to the estimated number of endmembers, which is obtained by HySime [76].

The solving method of SMoSU is based on MOEA/D, and the Tchebycheff decomposition was improved by adding the SID to help finding final solution, which can be written as

$$\begin{aligned} \min_{s_i} \quad & g_i^{te}(s_i|\lambda, z^*, s^*) = \max_{1 \leq j \leq 2} \left\{ \lambda_i^j |f_j(s_i) - z_i^*| \right\} \\ & \qquad + \mu \cdot \text{SID}(A_{s_i}, A_{s^*}) \\ \text{s.t.} \quad & 0 \leq \|s_i\|_0 \leq 2k \end{aligned} \tag{11}$$

where i is the ith subproblem, j is the jth objective function of MOP, z^* is a reference point that is the closest one to the origin in the population, and s^* is binary solution vector corresponding to z^*. A_{s_i} and A_{s^*} are the endmember matrixes corresponding to s_i and s^*, respectively. μ is the regularization parameter.

SMoSU is able to achieve good unmixing performance with a faster solving speed. However, there are some factors that limit its unmixing performance. First, the unmixing process considers only reconstruction error and sparsity, and the spatial information contained in hyperspectral image is ignored. Second, SMoSU measures the spectral similarity only by SID, so the magnitude and the angle of spectral are not considered. Finally, the generation of new individual depends only on the randomly flipping bits strategy without recombination, which makes it difficult to maintain population diversity, especially when the length of individual increases.

III. Proposed GMoSU

In this section, to solve the aforementioned problems of SMoSU, GMoSU will be proposed. The model that GMoSU incorporates spatial information as a multiobjective problem is described first, thus minimizing the error resulted by the convexification of the L_0-norm, and avoiding the difficulty about the selection of regularization parameters.

Specifically, a weighted graph G is generated to represent the hyperspectral image. Each vertex is a pixel and each edge connects similar vertices. Let $W \in R^{N \times N}$ be the affinity matrix of the graph, which characterizes the similarity of different pixels. Due to the reality that most of the materials in a given scene are clustered in a small area, the neighboring pixels of the scene have similar compositions, and their abundance

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are also similar [62]. Moreover, similar pixels have similar distances in the data space. This kind of relationship can be described as

$$\frac{1}{2}\sum_{i=1}^{N}\sum_{j\in\eta}\|x_i - x_j\|_2^2 W_{ij} \tag{12}$$

where x_i is the abundance vector of the ith pixel, and η denotes a $(2r+1)\times(2r+1)$ region, that is, the neighborhood of x_i and $r=2$. The affinity matrix is assigned

$$W_{ij} = e^{-\frac{\|y_i - y_j\|_2^2}{\sigma}} \tag{13}$$

which is the heat kernel [69], where σ is the density parameter that is set to $1/3$. Apparently, the closer the pixels y_i and y_j are, the larger the value of W_{ij} is. If the pixel y_j is not in the neighborhood η of pixel y_i, then $W_{ij}=0$.

For the convenience of calculation, (12) is transformed as follows:

$$\begin{aligned}\frac{1}{2}\sum_{i=1}^{N}\sum_{j\in\eta}\|x_i - x_j\|_2^2 W_{ij} &= \sum_{i=1}^{N} x_i^T x_i D_{ii} - \sum_{i,j=1}^{N} x_i^T x_j W_{ij}\\ &= Tr\left(XDX^T\right) - Tr\left(XWX^T\right)\\ &= Tr\left(XLX^T\right)\end{aligned} \tag{14}$$

where $D_{ii} = \sum_j W_{ij}$, which is a diagonal matrix, and $L = D - W$ is the graph Laplacian matrix. Thus, spatial information can be incorporated into unmixing by solving the following problem:

$$\min_{X} \quad Tr\left(XLX^T\right). \tag{15}$$

Motivated by PORSS, the proposed MOP describes

$$\min_{s\in\{0,1\}^m} \quad F(s) = [f_1(s), f_2(s), f_3(s)] \tag{16}$$

where

$$\begin{aligned} f_1(s) &= \begin{cases} +\infty, & \|s\|_0 = 0 \quad or \quad \geq 2k \\ \|Y - A_s X_s\|_F, & \text{Otherwise} \end{cases}\\ f_2(s) &= \left|\|s\|_0 - k\right|\\ f_3(s) &= \alpha * Tr\left(X_s * L * X_s^T\right)\end{aligned} \tag{17}$$

where α is a scaling factor, here it is set to 2e−4. Both k and s have the same meaning as mentioned in Section II-C, in which coding length m equals to the number of spectra in the library. While the endmember matrix A_s is obtained, and corresponding abundance matrix X_s is calculated as

$$X_s = \left(A_s^T A_s\right)^{-1} A_s^T Y. \tag{18}$$

Finally, the optimal solution $s^\star$ is selected. Furthermore, the fraction abundance can be calculated by the nonnegative least squares algorithm

$$\begin{aligned}\hat{X} = \arg\min_{X} \quad & \|Y - A_{s^\star} X\|_F\\ \text{s.t.} \quad & X \geq 0.\end{aligned} \tag{19}$$

Inspired by some related work [44], a novel Tchebycheff approach under MOEA/D framework is proposed, which incorporates a mixed metric in order to enforce the population concentrating into a single solution, and meanwhile improve its robustness to high noise. The improved Tchebycheff formula is

$$\begin{aligned}\min_{s_i} \quad & g_i^{te}\left(s_i|\lambda, z^*, s^*\right) = \max_{1\leq j\leq 3}\left\{\lambda_i^j |f_j(s_i) - z_i^*|\right\}\\ & \qquad + \mu\cdot \mathrm{NcS}\left(A_{s_i}, A_{s^*}\right)\\ \text{s.t.} \quad & 0 \leq \|s_i\|_0 \leq 2k\end{aligned} \tag{20}$$

where NcS is the abbreviation of NED-cosSAM-SID. i is the ith subproblem, j is the jth objective function of MOP, z^* is a reference point which is the closest one to the origin in the population, and s^* is the binary solution vector corresponding to z^*. The mixed metric NED-cosSAM-SID consists of normalized Euclidean distance (NED), cosine of SAM (cosSAM), and SID, which is first proposed in [73] for representative spectral extraction. A_{s_i} and A_{s^*} are the endmember matrix corresponding to s_i and s^*, respectively. μ is the regularization parameter, and μ is set to 0.8 here.

Note that the adjacent subproblems obtained by the Tchebycheff decomposition should have the similar optimal solution; therefore, a subproblem could be optimized with the information owning by its neighbor subproblems. In order to explore better solutions and escape the local optima, in each iteration, we propose a local recombination strategy in the neighborhood of the individual and apply a bitwise mutation strategy to generate new solutions. In order to improve the efficiency of endmember selection, RMUSIC [76] is used to prune the spectral library. The procedures of the proposed GMoSU are shown in Algorithm 1.

There are some details that should be explained further.

A. NED-cosSAM-SID(A_{s_i}, A_{s^*})

Let $s = [s_1, \ldots, s_{|s|}]$ be a binary vector in the current population, $s^* = [s_1^*, \ldots, s_{|s^*|}^*]$ is also a binary vector corresponding to reference point in the current population, and A_{s_1} and A_{s_2} are two spectral signals with L-bands, where $s_1 \in s$, $s_2 \in s^*$.

The NED(A_{s_1}, A_{s_2}) can be written as

$$\mathrm{NED}\left(A_{s_1}, A_{s_2}\right) = \sqrt{\sum_{i=1}^{L}\left(N\left(A_{s_1}^i\right) - N\left(A_{s_2}^i\right)\right)^2} \tag{21}$$

where

$$\begin{aligned} N\left(A_{s_1}^i\right) &= A_{s_1}/\sqrt{\sum_{i=1}^{L}\left(N\left(A_{s_1}^i\right)\right)^2}\\ N\left(A_{s_2}^i\right) &= A_{s_2}/\sqrt{\sum_{i=1}^{L}\left(N\left(A_{s_2}^i\right)\right)^2}.\end{aligned} \tag{22}$$

If the NED(A_{s_1}, A_{s_2}) is larger, the similar degree between A_{s_1} and A_{s_2} is lower.

The SAM is

$$\begin{aligned} & d_{\mathrm{SAM}}\left(A_{s_1}, A_{s_2}\right)\\ & \quad = \arccos\left(\frac{\sum_{i=1}^{L} A_{s_1}^i \cdot A_{s_2}^i}{\left[\sum_{i=1}^{L}\left(A_{s_1}^i\right)^2\right]^{\frac{1}{2}} \times \left[\sum_{i=1}^{L}\left(A_{s_2}^i\right)^2\right]^{\frac{1}{2}}}\right).\end{aligned} \tag{23}$$

Adding 1 to the cosSAM can transform the range from $[-1, 1]$ in $[0, \pi]$ to $[0, 2]$, and then taking the reciprocal of it consistent

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Algorithm 1: The Algorithm of GMoSU

Input:
Hyperspectral image data Y; spectral library A
Output:
Estimated Abundance matrix X
Preprocessing:
1. Estimate the number of active endmember as $\hat{k}$ of Hyperspectral image by HySime;
2. Prune the spectral library A to a size m by RMUSIC.
Initialization:
3. Population $S = \{s_1, \ldots, s_P\}$; weight vector set $\Lambda = [\lambda_1, \ldots, \lambda_P]$; neighborhood size NS; indexes of each subproblem's neighbors $\{B_1, \ldots, B_P\}$; maximum iteration T.
Endmember Selection:
while $t < T$ **do**
  4. $t = t + 1$;
  5. Set $\mu = 0$ if $t > 0.9 * T$
  for $i = 1, 2, \ldots, P$ **do**
    6. Randomly select an individual s_r in B_i;
    7. Apply local recombination on s_i, s_r to generate an individual s_i';
    8. Apply bitwise mutation on s_i' to generate a new individual s_i''.
    9. Set $z^* = F(s_i'')$ if $\|F(s_i'')\|_2 < \|F(s^*)\|_2$
    for $j \in B_i$ **do**
      10. Set $s_j = s_i''$ and $F(s_j) = F(s_i'')$ if $g_i^{st}(s_i'' \mid \lambda_i, z^*, s^*) \le g_j^{st}(s_j|\lambda_j, z^*, s^*)$
    end
  end
end
11. Return the final solution set as $s^\star$ and obtain corresponding endmember matrix as A_{s^*}.
Abundance Estimation:
12. Using the nonnegative least squares algorithm to compute the fraction abundance:
$\hat{X} = argmin_{x \ge 0}\|Y - A_{s^*}X\|_F$

with NED, it can be written as

$$\mathrm{cosSAM}\left(A_{s_1}, A_{s_2}\right) = \frac{1}{\cos\left(d_{\mathrm{SAM}}\left(A_{s_1}, A_{s_2}\right)\right) + 1}. \quad (24)$$

The value of $\mathrm{cosSAM}(A_{s_1}, A_{s_2})$ is larger, and the similarity between A_{s_1} and A_{s_2} is lower.

The $\mathrm{SID}(A_{s_1}, A_{s_2})$ is as follows:

$$\mathrm{SID}\left(A_{s_1}, A_{s_2}\right) = -\sum_{i=1}^{L} P_1^i \ln\frac{P_1^i}{P_2^i} - \sum_{i=2}^{L} P_2^i \ln\frac{P_2^i}{P_1^i} \quad (25)$$

where

$$P_1^i = A_{s_1}^i \Big/ \sum_{i=1}^{L} A_{s_1}^i$$
$$P_2^i = A_{s_2}^i \Big/ \sum_{i=1}^{L} A_{s_2}^i. \quad (26)$$

The larger the SID, the lower the similarity. Finally, the mixed spectral measure NED-cosSAM-SID (NcS) is obtained by multiplying the aforementioned three indexes, i.e., NED-cosSAM-SID = NED × cosSAM × SID, which is generally sensitive to distinction of spectral, even some tiny variations [73]. In the following, NED-cosSAM-SID(A_{s_i}, A_{s^*}) for all endmember spectral of A_{s_i} and A_{s^*} can be written as a matrix:

$$\widetilde{\mathrm{NcS}}(A_{s_i}, A_{s^*}) = \begin{bmatrix} \mathrm{NcS}\left(A_{s_1}, A_{s_1^*}\right) & \cdots & \mathrm{NcS}\left(A_{s_1}, A_{s^*_{\|s^*\|_1}}\right) \\ \vdots & \ddots & \vdots \\ \mathrm{NcS}\left(A_{s_{\|s\|_1}}, A_{s_1^*}\right) & \cdots & \mathrm{NcS}\left(A_{s_{\|s\|_1}}, A_{s^*_{\|s^*\|_1}}\right) \end{bmatrix}. \quad (27)$$

And $\mathrm{NcS}(A_{s_i}, A_{s^*})$ can be defined as

$$\mathrm{NcS}\left(A_{s_i}, A_{s^*}\right) = \frac{1}{2}\left(\frac{\sum_{i=1}^{\|s\|_1}\left\|\widetilde{\mathrm{NcS}}_{i,:}\right\|_\infty}{\|s\|_1} + \frac{\sum_{i=1}^{\|s^*\|_1}\left\|\widetilde{\mathrm{NcS}}_{:,i}\right\|_\infty}{\|s^*\|_1}\right). \quad (28)$$

This index can be used to calculate the difference between two endmember spectral matrixes of two individuals in the population, whose larger value corresponds to the lower similarity.

B. Generation of New Individuals

For fitting the MOEA/D framework, the local recombination strategy is incorporated into the generation of new individuals, whose main procedure is the following. Assume that m is the code length of individuals in population. Let s_i be the binary vector solution of the ith subproblem from Tchebycheff decomposition, and B_i is the neighbor of s_i. A solution vector s_r is randomly selected in B_i first. Next we apply one-point recombination on the solutions s_i and s_r, randomly select a bit of s_i and the same bit of s_r, and exchange all elements before this bit. Then, using uniform recombination, we exchange each bit of the two vectors in probability $1/2$ to generate s_i' and s_r'. Finally, we apply bitwise mutation on s_i' in probability $1/m$ to generate new solution s_i''. Comparing s_i and s_i'', if s_i'' is better, replace s_i with s_i'', otherwise keeping s_i in iteration. In order not to affect the optimization of other subproblems, we only update s_i during this process, not s_r.

An example is presented here so as to illustrate the benefit of the recombination during the generation of new individuals. Let $s_i = [0, 0, 0, 0, 0]$ and $s_r = [1, 1, 1, 1, 1]$. If the optimal solution is $[1, 0, 1, 0, 1]$, we can see that the probability of obtaining it is $(1/5)^5$ when applying bitwise mutation only. By contrast, the probability is $(1/5) \times (1/2)^5$ due to one-point recombination with probability $1/5$ and uniform recombination with probability $(1/2)^5$. Note that this advantage will become more and more obvious as the code length increases.

C. Role of the Scaling Factor

As the reference point in (20) is defined as the closest one to the origin in each iteration, it is calculated based on

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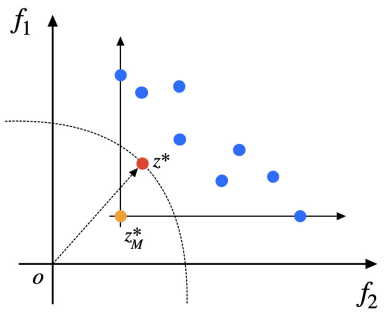


Fig. 1. Illustration about the reference point in GMoSU and the original one. All the blue points and the red point are individuals in population, and the red point z^* is the reference point in GMoSU, which is closest one to the origin in the population. The yellow point z_M^* is the original reference point in MOEA/D, which is a virtual point.

the selected endmembers. It is shown in Fig. 1. The yellow point z_M^* is the original reference point in MOEA/D, which is a virtual point actually. The red point denotes the reference point z^* in GMoSU, that is a real point, and more and more individuals will converge to it with iteration. In order to make the endmembers corresponding to the reference point not to deviate too much from the active endmembers, a scaling factor is integrated to the graph Laplacian objective function, which aims to balance it with other objectives. The introduction of the scaling factor ensures the correctness of endmember selection at the reference point. And its value is usually 2e−4, which will be confirmed by experiments.

IV. Experiments

In this section, we will conduct all experiments on synthetic and real hyperspectral image, and verify the proposed GMoSU in two aspects: its performance analysis and the comparison of different algorithms.

And we have considered two spectral libraries in our experiments.

1) $A_1 \in R^{224 \times 498}$: Selected from Chapter of Minerals in USGS digital spectral library(Splib06).[1] The library contains 498 spectra of different mineral materials under 224 bands distributed uniformly in 0.4–2.5 μm.
2) $A_2 \in R^{224 \times 201}$: Subset of A_1, where the spectral angle of any two different columns is larger than 5°. The spectral library has pruned because the same material in A_1 contains different spectral signatures with subtle changes.

Moreover, seven classic and state-of-the-art sparse unmixing algorithms are included in the comparison: SUnSAL [18], SUnSAL-TV [64], SMP [38], RSFoBa-2, RSFoBa-Inf [21], CSUnL0 [77], and SMoSU [44]. Specifically, SUnSAL is a classical convex relaxation unmixing method. SUnSAL-TV is an extension of SUnSAL, in which TV regularization is restricted to the assumption of local spatial similarity, and assumes that a pixel is only similar to its four neighbors. SMP, RSFoBa-2, RSFoBa-Inf belong to greedy approaches, in which RSFoBa-2 and RSFoBa-Inf contain the spatial-contextual regularized term based on graph Laplacian. CSUnL0 is a collaborative sparse unmixing approach and SMoSU is a multiobjective unmixing method recently. Note that the parameters of SUnSAL, SUnSAL-TV, SMP, RSFoBa-Inf, RSFoBa-2 and CSUnL0 need to be constantly adjusted for different images and noise levels, but the parameters of SMoSU and GMoSU are fixed once they are determined in advance. And in the GMoSU, the population size is set as 120, the number of maximum iteration is set as 150, the neighborhood size is 15, and unless specifically stated, the library size m after pruning is 40 [19], [44]. To improve the reliability, the experiments are repeated for ten times and we use their average results as the final results.

Two kinds of evaluation metrics were employed in our experiments for quantitative analysis. First, the signal-to-reconstruction error (SRE) was used to evaluate the unmixing accuracy, which is calculated as follows:

$$\mathrm{SRE} = 10\log_{10}\left(\frac{E\left[\|X\|_F^2\right]}{E\left[\|X-\hat{X}\|_F^2\right]}\right) \tag{29}$$

where X is the true abundance and $\hat{X}$ is the estimated abundance. For this criterion, the larger value of SRE means the higher accuracy. Furthermore, the root mean square error (RMSE) was used to measure the pixel reconstruction performance of the different methods, which is defined as

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{j=1}^{N}\|Y_j-\hat{Y}_j\|_2^2} \tag{30}$$

where Y is observed image without noise, and $\hat{Y}$ is reconstructed image, by which a lower value of RMSE means the better pixel reconstruction performance.

A. Performance of GMoSU

All experiments in this section mainly study its parameters analysis, robustness analysis for different numbers of endmembers with different noise level, and the ablation study of different strategies. The spectral library A_2 is used for this section in order to prevent the influence of the high correlation of the library. And ten spectra from the library A_2 are randomly chosen to generate homogeneous synthetic data, which include the rich spatial contextual information. These data will be used in parameters analysis and robustness analysis, and their generation process can be described as: 1) a scene of size 64 × 64 is divided into 8 × 8 sub regions; 2) each region is only filled with the spectral signal of the same type of endmembers, which are selected from the ten endmembers; 3) generating mixed pixels using an ideal low-pass filter; and 4) adding Gaussian white noise to generate synthetic image based on LMM.

1) Parameter Analysis: There are some parameters involved in GMoSU, such as, scaling factor α of f_3 in (17), neighborhood length r of pixel x_i, density parameter σ in weight matrix W and the weight coefficient μ of $\mathrm{NcS}(A_s, A_{s^*})$. In this part, we will discuss how to determine their optimal values. To this end, four spectra are randomly selected to generate homogeneous synthetic data according to the aforementioned

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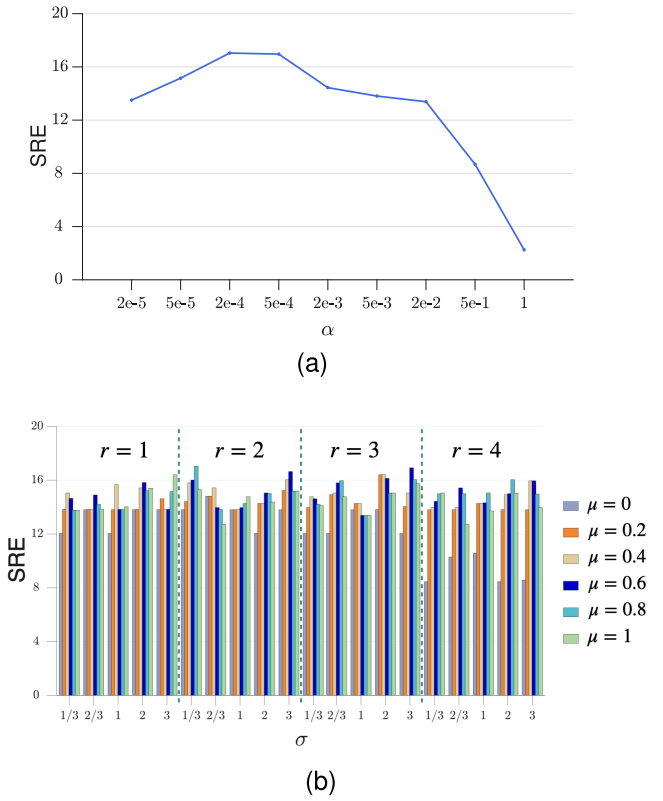


Fig. 2. (a) SRE obtained by GMoSU as a function of different parameter α. (b) SRE obtained by GMoSU as a function of different parameter r, μ and σ.

generation process, which are Hectorite SHCa-1, Carnallite HS430.3B, Diopside HS317.3B(Cr), and Hornblende_Mg NMNH117329. The SNR of Gaussian white noise is set as 20 dB. Fig. 2 depicts the SRE results for estimated abundances obtained by the proposed GMoSU under the different combinations of values of parameters. Specifically, Fig. 2(a) illustrates the influence of parameter α, from which we can see that an optimal performance can be obtained when the α equals to 2e$-$4. However, Fig. 2(b) depicts the SRE results of GMoSU under different combinations of values of r, μ and σ. It can be observed from Fig. 2(b) that higher SRE results can be obtained when $r = 2$, $\sigma = 1/3$ and $\mu = 0.8$.

It is worth noting that the reference point z^* is determined by its corresponding endmembers. If the selected endmembers are wrong, then the reference point will inevitably deviate from active endmembers with the population evolution. In the following, we first record the endmembers corresponding to the reference point when the iteration is made to the 1st, 30th, 60th, 90th and 120th times under the scaling factor $\alpha = 2\text{e}{-4}$ or $\alpha = 1$, and then compare Precision and Recall about endmember selection for further analyzing the role of the scaling factor. Here Precision measures the similarity between the selected endmembers and the active endmembers, and Recall measures the proportion of the selected active endmembers. Both of them are closer to 1, and the result of endmember selection is better. Table I shows the comparison results. We find that the Recall is just 1/2, and the Precision is reduced from 2/3 to 1/2 with iteration if $\alpha = 1$, i.e., not

TABLE I
PRECISION AND RECALL OF ENDMEMBER SELECTION

$\alpha = 2e-4$			$\alpha = 1$		
iterates	*Precision*	*Recall*	iterates	*Precision*	*Recall*
1	0	0	1	0	0
30	4/7	1	30	2/3	1/2
60	4/7	1	60	1/2	1/2
90	4/7	1	90	1/2	1/2
120	4/7	1	120	1/2	1/2

using the scaling factor α during endmember selection; on the contrary, the Recall is 1, and the Precision is 4/7 while setting $\alpha = 2\text{e}{-4}$, which means that when adding scaling factor, all active endmembers are correctly selected; meanwhile, virtual endmembers are reduced.

2) Robustness Analysis to Different Endmember Number and Noise Level: In this section, GMoSU will be compared with other sparse unmixing algorithms, such as SUnSAL, SUnSAL-TV, RSFoBa-Inf, RSFoBa-2, CSUnL0, and SMoSU. And the number of endmembers varies from 5 to 10, and we impose 20/30/40 dB Gaussian white noise to synthetic data, respectively. Table II lists all results with different endmember numbers and noise levels. Generally speaking, the SRE values of all algorithms are increased with the decrease of the noise level. And, since some algorithms actually work as an endmember selection method, once the exact endmembers are determined, their SREs should be the same. Specifically, in the low-noise conditions (30/40 dB), most of the methods perform well, especially GMoSU and SMoSU. However, in the high-noise level (20 dB), GMoSU behaves best in most cases, especially for the case of larger endmember number. These results indicate the superiority of GMoSU when incorporating the spatial information described by the graph Laplacian and the spectral characterization presented by the mixed metric into the proposed multiobjective optimization method.

3) Ablation Study: We will conduct ablation studies to verify the effectiveness of different key components of the proposed method, i.e., the graph Laplacian term in the objective function, the local recombination strategy, and the mixed spectral similarity metric. Specifically, ablation studies are conducted on four synthetic data, which are named A, B, C, and D, respectively. In the first three data, as used earlier, the number of endmembers varies from 4 to 6. As to the fourth data, its abundance is generated according to [77], and five endmembers are randomly selected from the library A_2 and included, such as Epsomite GDS149, Clinochlore_Fe SC-CCa-1.c, Clinoptilolite GDS152, Gypsum SU2202, and Heulandite NMNH84534. All the data are corrupted by 20 dB Gaussian white noise. Since the parameters jointly affect the performance of the proposed method, the parameters are also tuned jointly when conducting ablation studies.

a) Effectiveness with or without the graph Laplacian: We removed the graph Laplacian term from the objective function of our method, and then compared the SRE of unmixing of GMoSU and GMoSU without the graph Laplacian. Additionally, GMoSU can be regarded as an extension of

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TABLE II
SRE RESULTS OF SUnSAL, SUnSAL-TV, SMP, RSFoBa-Inf, RSFoBa-2, CSUnL0, SMoSU, AND GMoSU

SNR	k	SUnSAL	SUnSAL-TV	SMP	RSFoBa-Inf	RSFoBa-2	CSUnL0	SMoSU	GMoSU
20dB	5	1.9766	4.3098	5.0652	7.3811	5.2849	11.3793	13.3008	**14.6462**
	6	1.7016	4.7485	3.4759	3.5124	6.2399	10.6485	9.0701	**12.5295**
	7	3.0238	5.0085	4.5464	4.5648	5.7859	6.9199	6.3340	**12.1472**
	8	2.4812	6.1685	2.7740	3.9804	3.7895	4.2031	6.0646	**8.7892**
	9	1.8891	3.9417	1.7363	1.7988	2.9767	3.6815	5.7779	**9.9119**
	10	1.8624	4.9457	1.4355	2.4927	2.0342	2.7179	6.3703	**8.1816**
30dB	5	5.5835	10.2013	15.8478	19.4920	22.0462	24.8703	**25.5815**	**25.5815**
	6	5.5354	12.1069	11.8121	17.3819	16.3722	23.3724	**23.9828**	**23.9828**
	7	6.4426	11.0150	15.5595	18.3679	18.8226	24.1308	23.8065	**24.4092**
	8	6.4158	11.2598	14.5394	14.5461	16.0417	12.1493	**19.5879**	**19.5879**
	9	5.4294	8.3993	14.0160	12.8407	14.1772	12.2962	16.8596	**19.0787**
	10	6.7282	9.8544	13.7197	12.8760	10.8276	11.8567	15.7163	**18.5829**
40dB	5	10.0966	17.0303	20.1768	26.4071	31.7664	35.1950	**35.4312**	**35.4312**
	6	10.3076	20.3767	22.3615	27.6309	29.0340	33.6378	**33.7977**	**33.7977**
	7	11.3509	18.1545	21.9698	25.1587	23.4923	34.0474	**34.1158**	**34.1158**
	8	11.6136	17.7731	23.4605	23.8455	25.3261	28.3573	**28.4150**	**28.4150**
	9	10.4978	15.3445	21.0670	22.5511	23.3633	28.2536	**28.2761**	**28.2761**
	10	10.8223	15.3973	19.4002	22.8505	19.3151	27.0464	**27.0858**	**27.0858**

TABLE III
SRE RESULTS OF GMoSU WITH OR WITHOUT THE GRAPH LAPLACIAN

Strategies	data A	data B	data C	data D
SMoSU	11.5163	13.3008	9.0701	7.5661
w.o. the graph Laplacian	14.2086	13.4225	10.6609	17.5546
w. the graph Laplacian	**17.0403**	**14.6462**	**12.5295**	**19.6928**

TABLE IV
SRE RESULTS OF GMoSU AND GMoSU WITHOUT THE LOCAL RECOMBINATION STRATEGY

Strategies	library size	data A	data B	data C	data D
w.o. the local recombination strategy	40	16.5353	**14.9023**	**12.7021**	19.04771
	80	12.8971	11.7905	9.7981	17.7129
	150	8.7218	7.3649	5.8093	8.8160
	201	3.5746	2.6487	2.6048	5.3008
w. the local recombination strategy	40	**17.0403**	14.6462	12.5295	**19.6928**
	80	**16.8198**	**14.0523**	**11.2547**	**18.3993**
	150	**16.4750**	**14.2780**	**11.3100**	**18.0418**
	201	**14.4832**	**13.0675**	**11.2288**	**17.4861**

SMoSU. When $\alpha = 0$ in (17), f_3 will be omitted; thus the spatial information will not be leveraged. Then GMoSU is similar to SMoSU except that SMoSU doesn't involve the local recombination strategy and the mixed spectral similarity metric. So we also compared the unmixing results of GMoSU with and without graph Laplacian to that of SMoSU in this part. Obtained results by GMoSU, GMoSU without graph Laplacian, and SMoSU were summarized in Table III. Here, the parameter μ in the GMoSU without the graph Laplacian term was set to 0.8. Regarding SRE, GMoSU performs best, which is followed by GMoSU without the graph Laplacian and then SMoSU, which preliminarily proves that the graph Laplacian is relevant and effective. Fig. 3 presented the estimated abundance maps of the synthetic data D. It shows that the GMoSU and GMoSU without the graph Laplacian have selected all active endmembers correctly, and the maps obtained by GMoSU contain fewer noise points compared to GMoSU without the graph Laplacian. However, only three active endmembers are selected by SMoSU. To some extent, these results further verify the effectiveness of the graph Laplacian term and GMoSU's superiority over SMoSU.

b) Effectiveness with or without the local recombination strategy: In order to confirm the active role that the local recombination strategy plays in GMoSU, we performed ablation study regarding it based on library with different size, respectively. Here, the number of endmember spectra retained by the library A_2 after pruning was 40, 80, 150, and 201(namely no pruning), respectively. Table IV summarizes the unmixing results with libraries of different sizes. It can be seen from Table IV that the performance of the method decreases considerably when the local recombination strategy was removed, especially when the number of spectra in the library increases. This is because the local recombination strategy can reduce the influence of large spectral library size, improve the species' diversity, and escape the local optima. Thus, these results confirm that local recombination strategy indeed played significant role in GMoSU.

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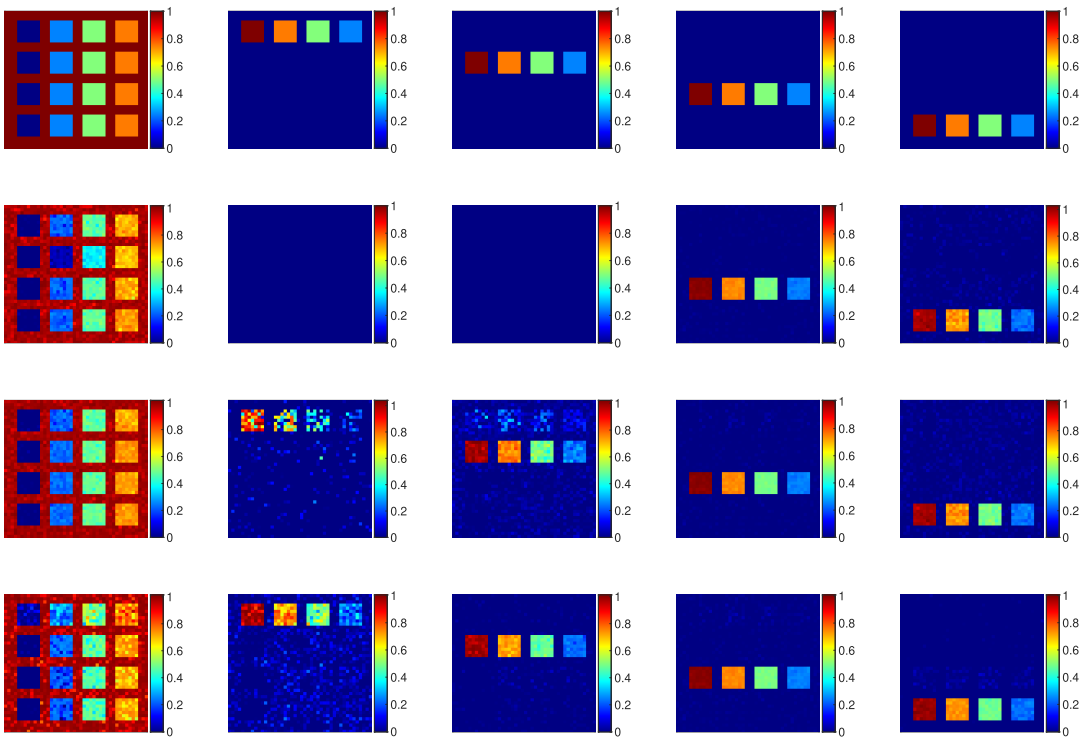


Fig. 3. Abundance maps of five endmembers. (Top row to bottom row) Maps obtained by the truth, SMoSU GMoSU without the graph Laplacian and GMoSU. (Left column to right column) Maps corresponding to Epsomite GDS149, Clinochlore_Fe SC-CCa-1.c, Clinoptilolite GDS152, Gypsum SU2202, and Heulandite NMNH84534, respectively.

TABLE V

SRE RESULTS OF GMoSU WITH OR WITHOUT THE MIXED SPECTRAL SIMILARITY METRIC REGULARIZATION NED-COSSAM-SID

Strategies	data A	data B	data C	data D
w.o. the mixed spectral similarity metric	14.6324	12.9522	8.4526	13.2159
w. the mixed spectral similarity metric	**17.0403**	**14.6462**	**12.5295**	**19.6928**

c) Effectiveness with or without the mixed spectral similarity metric: In this part, we investigated the role of the mixed spectral similarity metric regularization NED-cosSAM-SID of (20). The parameter σ and r were set to 2/3 and 2 for better results in the method without the mixed spectral similarity metric. Table V summarizes the unmixing performances of GMoSU employing the Chebycheff formula with or without the mixed spectral similarity metric regularization. As can be seen, when our method employed the mixed spectral similarity metric regularization, the higher SRE was obtained. Therefore, these results imply that the Chebycheff formula with the NED-cosSAM-SID regularization plays marked role in GmoSU, since it makes the final solution be much closer to the actual endmember matrix in the hyperspectral image.

B. Comparison of Different Algorithms

In this section, two synthetic data and one real data are used to compare our method with other sparse unmixing algorithms. We conduct these experiments on the library A_1.

1) Synthetic Data I: These data are a 64 $\times$ 64 synthetic hyperspectral image with a well spatial-contextual homogeneity, which is designed by Tang et al. [14]. And its abundance is limited to smaller than 0.7 to avoid pure pixels. Moreover, five active endmembers are randomly selected in library, they are Chromite HS281.3B, Colemanite GDS143, Copiapite GDS21, Hypersthene PYX02.e 34 μm, and Lizardite NMNHR4687.a 280, respectively. The results were shown in Table VI and Fig. 4. Table VI listed all SREs and RMSEs of all methods. Multiobjective-based methods, namely SMoSU and GMoSU, significantly outperforms other methods, which have the highest SREs and the lower RMSEs. Note that GMoSU achieves the same performance as SMoSU under the higher SNR (30/40 dB), but it is the best one in 20 dB. Table VII recorded the execution times of the proposed method and comparison methods. Here the execution time refers to the required iteration time until the solutions keep stable within ten consecutive iterations. It was observed that the execution time of the proposed GMoSU is longer than SMoSU, since the proposed method includes the graph Laplacian, the local recombination strategy, and the mixed spectral similarity metric. Under the higher SNRs, the results of SMoSU and GMoSU are similar,but SMoSU's solving speed is faster. In Fig. 4, the abundance maps of these methods are presented in 30 dB Gaussian white noise. From left column to right column are the abundance maps obtained by SUnSAL, SUnSAL-TV, SMP, RSFoBa-Inf, RSFoBa-2, CSUnL0, SMoSU, GmoSU, and the truth. From top row to bottom row are the abundance maps corresponding to Goethite WS219 (limonite), Hematite GDS69.b 104-150u, Ammonio-jarosite SCR-NHJ, Cinnabar

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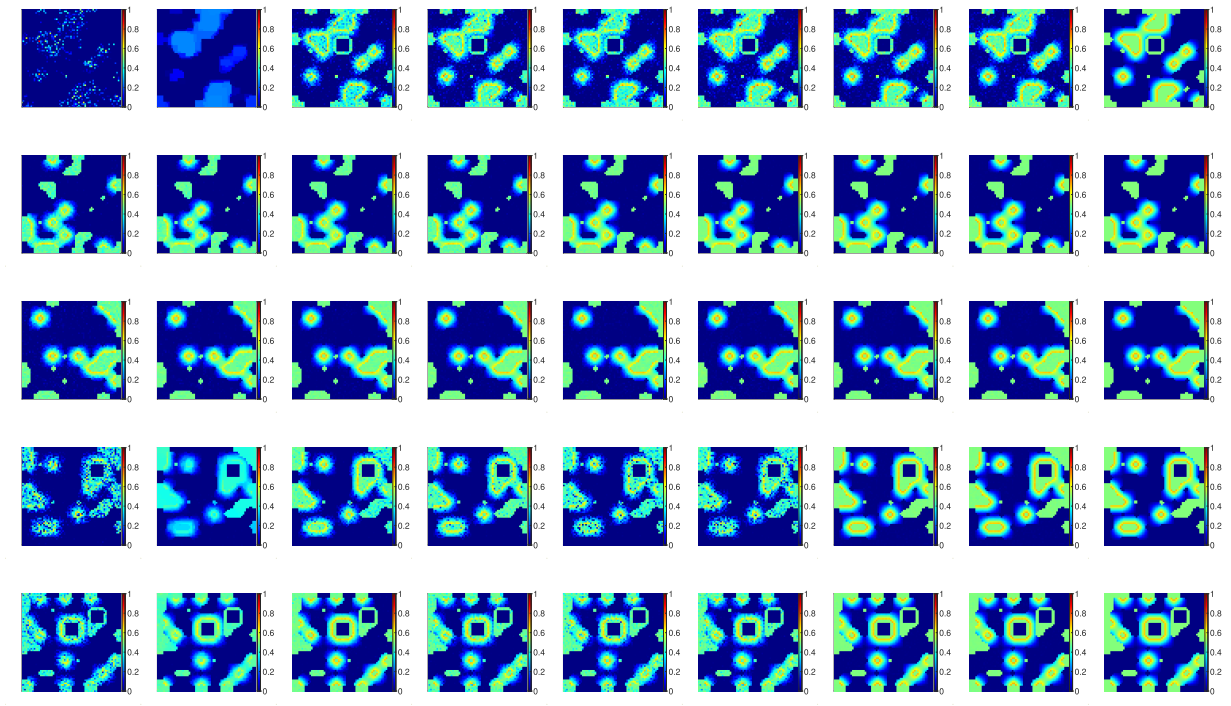


Fig. 4. Abundance maps of synthetic data I with 30 dB Gaussian white noise. (Left column to right column) Abundance maps obtained by SUnSAL, SUnSAL-TV, SMP, RSFoBa-Inf, RSFoBa-2, CSUnL0, SMoSU, GmoSU, and the truth. (Top row to bottom row) Maps corresponding to Goethite WS219 (limonite), Hematite GDS69.b 104-150u, Ammonio-jarosite SCR-NHJ, Cinnabar HS133.3B, and Europium_Oxide GDS33.

TABLE VI

SRE/RMSE RESULTS OF SUnSAL, SUnSAL-TV, SMP, RSFoBa-Inf, RSFoBa-2, CSUnL0, SMoSU, AND GMoSU ON SYNTHETIC DATA I

SNR	SUnSAL	SUnSAL-TV	SMP	RSFoBa-Inf	RSFoBa-2	CSUnL0	SMoSU	GMoSU
20	2.8928/0.4623 ($\lambda = 0.03$)	5.6318/0.1191 ($\lambda = 5e-3$, $\lambda_{TV} = 5e-3$)	5.0308/0.1003 ($Block.Size = 25$, $t = 0.7$)	5.9780/0.0991 ($Block.Size = 10$, $\lambda = 5e-3$)	4.4630/0.1014 ($Block.Size = 10$, $\lambda = 0.2$)	6.6850/0.0928 ($a = 0.06$)	9.2941/0.0785	**12.9402/0.0781**
30	5.2400/0.5777 ($\lambda = 4e-4$)	8.5816/0.0462 ($\lambda = 0$, $\lambda_{TV} = 5e-4$)	14.9033/0.0327 ($Block.Size = 28$, $t = 0.85$)	14.5927/0.0316 ($Block.Size = 10$, $\lambda = 0.05$)	12.2357/0.0319 ($Block.Size = 15$, $\lambda = 0.3$)	12.0407/0.0274 ($a = 6e-4$)	**23.4489/0.0232**	**23.4489/0.0232**
40	10.2966/0.3295 ($\lambda = 1e-4$)	12.5901/0.0150 ($\lambda = 0$, $\lambda_{TV} = 1e-4$)	25.3863/0.0101 ($Block.Size = 32$, $t = 0.96$)	24.9757/0.0095 ($Block.Size = 10$, $\lambda = 0.02$)	21.6086/0.0103 ($Block.Size = 15$, $\lambda = 0.2$)	19.4009/0.0090 ($a = 8e-5$)	**33.2826/0.0074**	**33.2826/0.0074**

TABLE VII

TIME(S) OF SUnSAL, SUnSAL-TV, SMP, RSFoBa-Inf, RSFoBa-2, CSUnL0, SMoSU, AND GMoSU ON SYNTHETIC DATA I

SNR	SUnSAL	SUnSAL-TV	SMP	RSFoBa-Inf	RSFoBa-2	CSUnL0	SMoSU	GMoSU
20	6.98	83.95	4.06	3.02	4.41	127.32	87.82	126.17
30	6.84	84.08	4.69	3.61	5.08	178.01	81.26	123.68
40	5.93	86.14	4.71	3.11	5.29	169.06	82.06	123.87

HS133.3B, and Europium_Oxide GDS33. It can be observed that all methods cannot recognize Goethite WS219 (limonite) well, neither SUnSAL nor SUnSAL-TV even reconstructs its fractional abundance. As for other methods, they also contain significant noise in fractional abundance of Goethite WS219 (limonite). For abundance maps of Cinnabar HS133.3B, the reconstructing results of SUnSAL and SUnSAL-TV are not well, the results corresponding to SMP, RSFoBa-Inf, and RSFoBa-2 also contain many noise; however, the abundance maps obtained by SMoSU and GMoSU are clearer, without obvious noise. In general, GMoSU and SMoSU obtain better visual results and contain fewer noise points.

2) Synthetic Data II: These data are generated by a Hyperspectral Imagery Synthesis toolbox.[2] To increase the unmixing

[2]Available online: http://www.ehu.es/ccwintco/index.php

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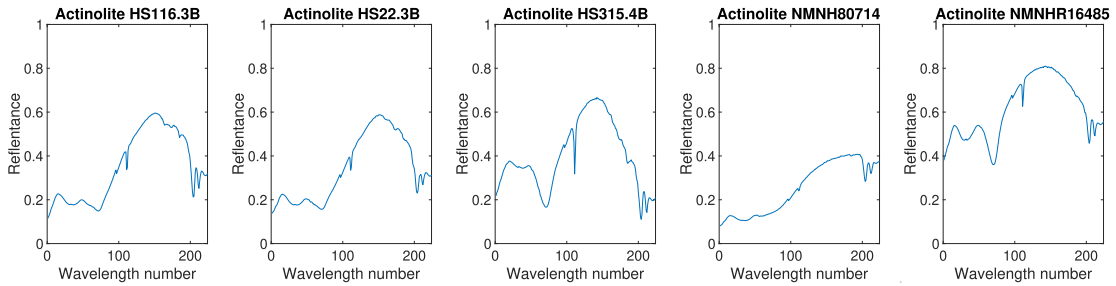


Fig. 5. Spectra of five endmembers in synthetic data II.

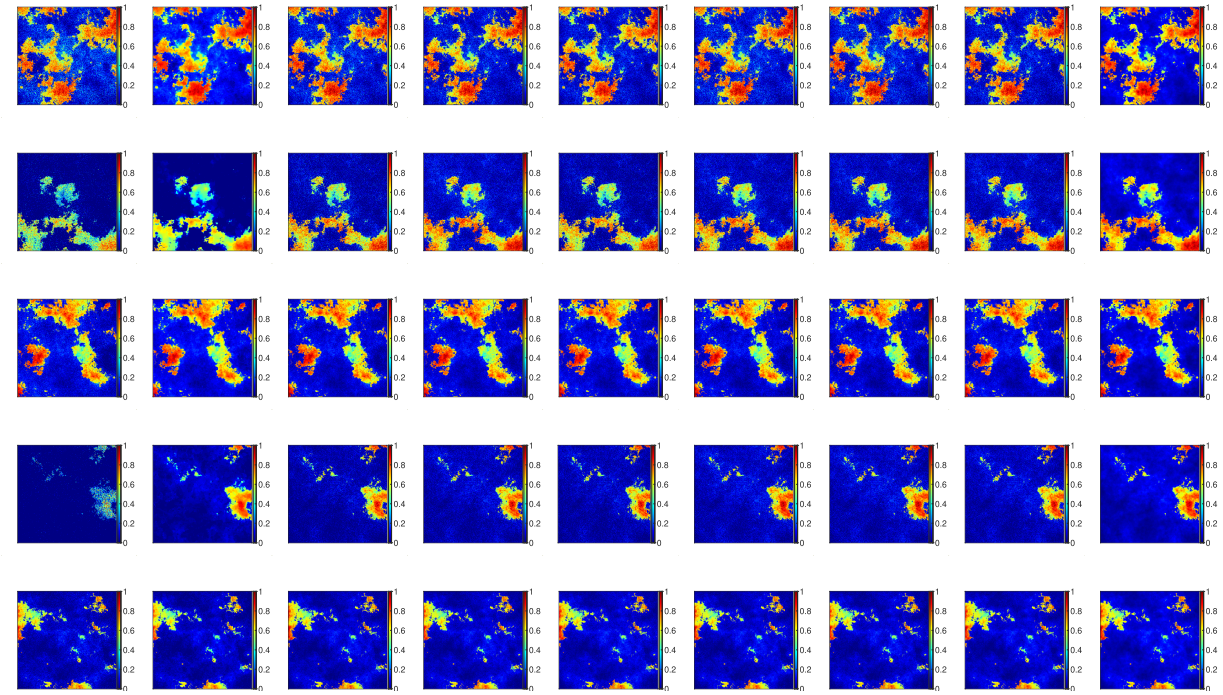


Fig. 6. Abundance maps of synthetic data II with 30 dB Gaussian white noise. (Left column to right column) Abundance maps obtained by SUnSAL, SUnSAL-TV, SMP, RSFoBa-Inf, RSFoBa-2, CSUnL0, SMoSU, GmoSU, and the truth. (Top row to bottom row) Maps corresponding to Actinolite HS116.3B, Actinolite HS22.3, Actinolite HS315.4B, Actinolite NMNH80714, and Actinolite NMNHR16485.

TABLE VIII

SRE/RMSE RESULTS OF SUnSAL, SUnSAL-TV, SMP, RSFoBa-Inf, RSFoBa-2, CSUnL0, SMoSU, AND GMoSU ON SYNTHETIC DATA II

SNR	SUnSAL	SUnSAL-TV	SMP	RSFoBa-Inf	RSFoBa-2	CSUnL0	SMoSU	GMoSU
20	4.1307/0.8317 ($\lambda = 2e-2$)	9.0283/0.1291 ($\lambda = 1e-4$, $\lambda_{TV} = 5e-3$)	5.6652/0.1569 ($Block.Size = 128$, $t = 0.96$)	7.7005/0.0832 ($Block.Size = 42$, $\lambda = 0.3$)	7.2864/0.0833 ($Block.Size = 42$, $\lambda = 1.2$)	9.0262/0.0796 ($a = 0.01$)	7.8224/0.0844	**9.0777/0.0790**
30	9.0711/0.5838 ($\lambda = 2e-3$)	14.0564/0.0474 ($\lambda = 0$, $\lambda_{TV} = 5e-4$)	14.4128/0.0336 ($Block.Size = 128$, $t = 0.96$)	15.6393/0.0285 ($Block.Size = 42$, $\lambda = 0.3$)	14.6078/0.0292 ($Block.Size = 30$, $\lambda = 1.2$)	15.4343/0.0318 ($a = 0.01$)	14.2538/0.0304	**15.9733/0.0277**
40	14.7191/0.0019 ($\lambda = 1e-4$)	20.8830/0.0139 ($\lambda = 0$, $\lambda_{TV} = 3e-4$)	20.8830/0.0115 ($Block.Size = 128$, $t = 0.96$)	**24.8997/0.0092** ($Block.Size = 45$, $\lambda = 0.3$)	22.3698/0.0100 ($Block.Size = 30$, $\lambda = 1.2$)	22.9285/0.0103 ($a = 0.008$)	24.6695/0.0096	**24.8997/0.0092**

difficulty, five signatures with similar spectra are selected from the library. They are Actinolite HS116.3B, Actinolite HS22.3, Actinolite HS315.4B, Actinolite NMNH80714, and Actinolite NMNHR16485, whose spectra are shown in Fig. 5. A size of 256 × 256 image is generated using these endmembers according to the Spheric Gaussian Field, which contains rich spatial-contextual information and complex generally. Note that during the process of the image generation, we use the default parameters provided in the toolbox. Table VIII presented all SREs and RMSEs results, and we found that the unmixing performance and the pixel reconstruction performance of GMoSU is better in all noise levels. However, the execution time of different methods were shown in Table IX. And Fig. 6 gives the abundance maps of these methods

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TABLE IX
TIME(S) OF SUNSAL, SUNSAL-TV, SMP, RSFOBA-INF, RSFOBA-2, CSUNL0, SMOSU, AND GMOSU ON SYNTHETIC DATA II

SNR	SUnSAL	SUnSAL-TV	SMP	RSFoBa-Inf	RSFoBa-2	CSUnL0	SMoSU	GMoSU
20	183.17	2307.98	63.39	122.82	178.53	4207.94	183.12	267.19
30	190.37	1921.26	67.93	128.13	60.68	561.26	293.26	316.05
40	146.47	1731.36	78.37	147.18	61.61	588.49	166.51	242.23

TABLE X
RMSE RESULTS OF SUNSAL, SUNSAL-TV, SMP, RSFOBA-INF, RSFOBA-2, CSUNL0, SMOSU, AND GMOSU ON REAL DATA

SUnSAL	SUnSAL-TV	SMP	RSFoBa-Inf	RSFoBa-2	CSUnL0	SMoSU	GMoSU
1.1264	0.1096	0.1148	0.1112	0.1102	0.1035	0.1065	**0.1012**

TABLE XI
TIME(S) OF SUNSAL, SUNSAL-TV, SMP, RSFOBA-INF, RSFOBA-2, CSUNL0, SMOSU, AND GMOSU ON REAL DATA

SUnSAL	SUnSAL-TV	SMP	RSFoBa-Inf	RSFoBa-2	CSUnL0	SMoSU	GMoSU
124.22	2055.22	229.91	198.33	210.33	3217.42	625.89	828.93

in 30 dB Gaussian white noise, and the abundance maps obtained by SUnSAL-TV show better spatial smoothness, and meanwhile, CSUnL0, SmoSU, and GMoSU present clearer abundance maps than others.

3) Real Data: Fig. 7 shows the mineral distribution produced by Tricorder 3.3 software, where each pixel is considered as a pure pixel and is labeled as a special material which is the most representative. Since the Tricorder map was produced in 1995, but the publicly available AVIRIS Cuprite data[3] were collected in 1997. However, it is almost impossible to obtain the true abundances of different materials in real hyperspectral data. Therefore, we only compared the pixel reconstruction errors to make a quantitative analysis and used the minerals map as a reference to make a qualitative analysis. Specifically, a 250 × 191 pixels subset of the AVIRIS Cuprite data is adopted, which is shown using a red box in Fig. 7. The considered scene only contains 188 spectral bands, since water absorption bands and some low SNR bands (1–2, 105–115, 150–170, and 223–224) are already removed in advance. And the corresponding bands of the spectral library A_1 are also removed before unmixing. Moreover, the parameters of all algorithms are appropriately adjusted based on their original works, respectively, such as, $\lambda = \lambda_{\mathrm{TV}} = 1\mathrm{e}{-4}$ for SUnSAL and SUnSAL-TV, Block.Size = 20, $t = 0.96$ for SMP, Block.Size = 15, $\lambda = 0.05$ for RSFoBa-Inf, Block.Size = 15, $\lambda = 0.1$ for RSFoBa-2, $a = 6e-5$ for CSUnL0. Note that we estimate the number of endmembers to be 16 based on HySime in this subscene; the whole endmember number is set larger than 16; and the library size m after pruning is set as 180 to ensure that the active endmembers should not be missed in RMUSIC. The E-3DTV method [78] was used to realize

[3]Available online: http://aviris.jpl.nasa.gov/html/aviris.freedata.html

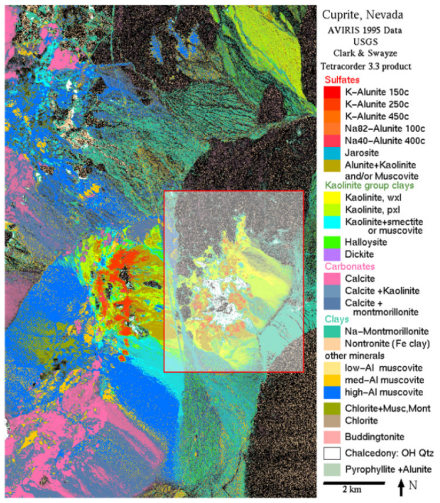


Fig. 7. Material distribution map of Curprite mining from Tricorder 3.3 software.

the hyperspectral denoising before the quantitative analysis. Table X listed the RMSE results of the proposed GMoSU and other unmixing algorithms. As can be seen, the proposed GMoSU outperforms the other comparison algorithms with respect to pixel reconstruction performance.

Fig. 8 shows the estimated abundance maps of the minerals: Alunite + Kaolinite/Muscovite, Buddingtonite, Hematite, and Montmorillonite. From left column to right column are the abundance maps obtained by SUnSAL, SUnSAL-TV, SMP, RSFoBa-Inf, RSFoBa-2, CSUnL0, SMoSU, and GMoSU. From top row to bottom row are the abundance maps corresponding to the minerals: Alunite + Kaolinite/Muscovite, Buddingtonite, Hematite, and Montmorillonite. We can see

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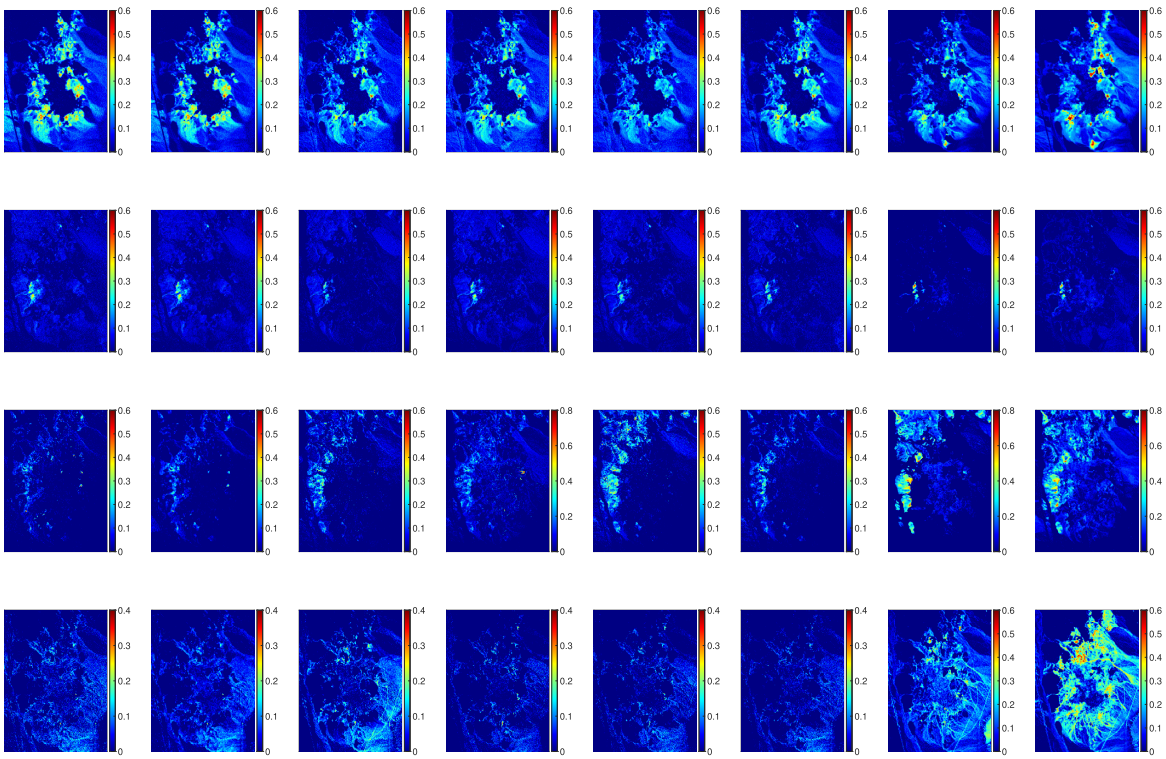


Fig. 8. Estimated abundance maps of Cuprite data. (Left column to right column) Estimated abundance maps obtained by SUnSAL, SUnSAL-TV, SMP, RSFoBa-Inf, RSFoBa-2, CSUnL0, SmoSU, and GMoSU. (Top row to bottom row) Maps corresponding to Alunite+Kaolinite/Muscovite, Buddingtonite, Hematite, and Montmorillonite.

that the estimated abundances by GMoSU show better visual results, especially the Montmorillonite mineral. For the Hematite mineral, SUnSAL, SUnSAL-TV, and RSFoBa-Inf give blurred abundance maps. In general, the proposed GMoSU has ability to obtain more clear abundance maps of displayed minerals compared with several existing methods. Table XI summarized the execution times of all methods. It can be seen from the table that the proposed GMoSU could obtain better pixel reconstruction performance and better visual results, at the cost of longer execution time.

V. Conclusion

In this work, we propose a GMoSU, which is motivated by the fact that spatial characteristics have been successfully exploited in the single-objective unmixing method. We exploit the merit of the spatial information to improve the unmixing accuracy based on the multiobjective rather than the single-objective. First, a multiobjective subset selection-based framework is presented, in which the reconstruction error, the sparsity error, and the graph Laplacian are contained as three parallel objective functions, thus avoiding the selection of regularization parameters. In this model, the reconstruction error is based on the spectral signature, the sparsity is measured by L_0-norm, and spatial information is represented by the graph Laplacian, thus fully utilizing the internal information and minimizing the error resulted by the approximation of the L_0-norm. Furthermore, considering that the evolution algorithm should improve the species diversity, we propose a novel local recombination strategy to generate diverse solutions, which corresponds to the MOEA/D framework. Specifically, an individual is updated through recombining with another individual selected randomly, in which each individual is encoded as a binary vector, and the updating is proceeded only in the neighborhood other than the whole population. The local recombination could help evolution both escape the local optima easily and reduce the influence of large spectral library size. Moreover, to settle the problem of the final solution selection, a mixed spectral similarity metric is incorporated into Tchebycheff formula in order to fully reflect the spectral characterization, such as spectral amplitude, angle, and information entropy, which are measured by ED, SAM, and SID. Different from the general MOP, which strives to obtain a Pareto-optima set, GMoSU could enforce the population converge to a single solution exactly.

In the experimental part of this work, many simulations are conducted to verify the proposed GMoSU, such as the performance analysis and comparison experiments. Specifically, based on performance analysis experiments, we first analyze the selection of parameters involved in GMoSU, and find their optimal values for the best unmixing results, for example, the more suitable the scaling factor in the graph Laplacian is, the more exactly active endmembers are selected. Then, we test the robustness of GMoSU to different endmember numbers and noise levels, which indicate that proposed method is more robust compared with several existing algorithms, especially under high noise conditions. Moreover, as another proof of concept, the ablation studies regarding different key components of the proposed method were conducted to assess their effectiveness, respectively. Additionally, since GMoSU has the model similar to the SMoSU method when the graph Laplacian is omitted, we also compared GMoSU without the graph Laplacian to the SMoSU method; experiments show that

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it still performs better. All results of ablation studies verified the effectiveness of each components of the proposed method for improving the unmixing performance. In the following comparison experiments, two synthetic data and one real data are used for comparing GMoSU with several state-of-art methods. From these quantitative and qualitative results, we find that GMoSU also has ability to recognize nearly all active endmembers and reconstruct more clear abundance maps if the image contains spatial-contextual homogeneity. Due to introduced local recombination strategy and mixed spectral similarity metric, relatively the proposed GMoSU require more computation time. In most cases, the cost of more computation time was paid by superior unmixing performance. To this end, the main focus of future study is further accelerating the proposed GMoSU and accomplishing elevated balance between unmixing performance and execution time. In conclusion, through the good performance of the proposed GMoSU, we find that exploring more internal attributes could assist us to explore effective techniques for improving hyperspectral image recognition, which will be one of the development directions of the remote sensing technology.

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